10 types of **optimizers** used in **Machine Learning** with short explanation for each.

1. **Gradient Descent (GD)**: Iteratively updates model parameters to minimize loss by moving in the negative gradient direction.
2. **Stochastic Gradient Descent (SGD)**: A variant of GD that updates parameters using a randomly chosen subset (mini-batch) of data for faster convergence.
3. **Mini-batch Gradient Descent**: Combines GD and SGD by using small batches of data to balance efficiency and stability.
4. **Adagrad**: Adapts the learning rate for each parameter based on past gradients, benefiting sparse data.
5. **RMSprop**: Adjusts learning rates adaptively by dividing gradients by a moving average of recent magnitudes, suitable for non-stationary objectives.
6. **Adam (Adaptive Moment Estimation)**: Combines momentum and adaptive learning rates for efficient training and fast convergence.
7. **AdaDelta**: An extension of Adagrad that limits the accumulation of past gradients to avoid aggressive decreasing learning rates.
8. **Nadam**: A variant of Adam incorporating Nesterov momentum for potentially faster convergence.
9. **Momentum**: Accelerates SGD by adding a fraction of the previous update to the current update, improving convergence speed.
10. **Newton's Method**: Uses second-order derivatives (Hessian) to find parameters by approximating the loss surface curvature, often computationally expensive.